AWS VPC Peering Project

Configuration, Connectivity Testing, and Evidence Report

AWS Region: Europe (Stockholm) — eu-north-1

Objective: Establish private network connectivity between two VPCs using VPC Peering, configure routes in both VPCs, and validate access to a private EC2 instance.

1. Architecture Overview

VPC-A (10.0.0.0/16) contains a public subnet hosting Bastion-VPC-A and a private subnet hosting Private-Server-A. VPC-B (10.1.0.0/16) contains Private-Server-B in a private subnet. The VPCs are connected by the peering connection pcx-008dccff0cf5e2620.

VPC-A 10.0.0.0/16	**VPC Peering** pcx-008dccff0cf5e2620 Status: Active	**VPC-B** 10.1.0.0/16
Bastion: 10.0.1.124 Private server: 10.0.2.218	Routes in both directions	Private server: 10.1.1.81

2. Network Configuration

Resource	Configuration
VPC-A	vpc-05d79e45930f20c75 | 10.0.0.0/16
VPC-A public subnet	Public-Subnet-A | 10.0.1.0/24
VPC-A private subnet	Private-Subnet-A | 10.0.2.0/24
VPC-B	vpc-04aa68eeed7fa8d52 | 10.1.0.0/16
VPC-B private subnet	Private-Subnet-B | 10.1.1.0/24
Peering	Peering-A-to-B | pcx-008dccff0cf5e2620 | Active
EC2 private IPs	Bastion: 10.0.1.124 | Server-A: 10.0.2.218 | Server-B: 10.1.1.81

3. Routing

VPC-A route tables include a route for 10.1.0.0/16 targeting the peering connection. VPC-B's private route table includes the return route for 10.0.0.0/16 targeting the same connection. The VPC-A public route table also includes a default route to the Internet Gateway.

4. Connectivity Tests

Ping: Four packets were sent to 10.1.1.81; all four were received (0% packet loss), with average reported RTT of 0.179 ms.

SSH: SSH from Windows through Bastion-VPC-A to 10.1.1.81 succeeded and returned hostname `ip-10-1-1-81`.

5. Conclusion

The supplied AWS console screenshots and terminal output document an Active peering connection, configured routes in both VPCs, and successful ping/SSH tests to Private-Server-B. The evidence pages that follow are included as an appendix.

Security: Never include the PEM private key or AWS credentials in the report or source repository.

Appendix A — Evidence Screenshots

The following screenshots were supplied for the project evidence appendix.

Figure 1. VPC peering connection Peering-A-to-B shows Active status.

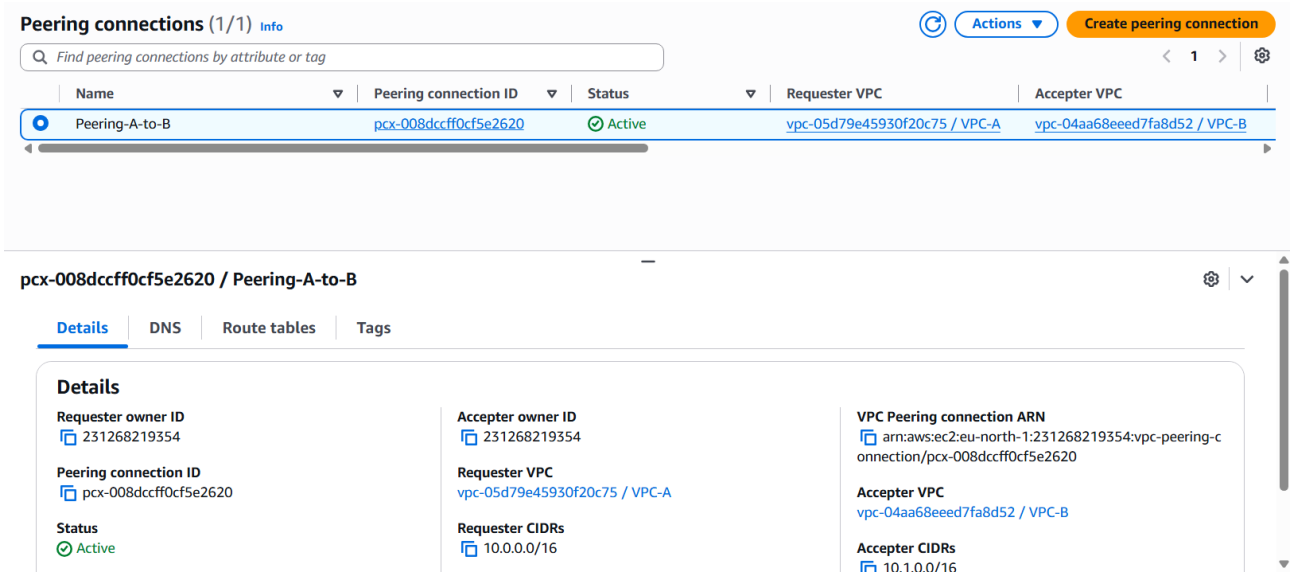


Figure 2. VPC-A public route table, including Internet Gateway and VPC peering routes.

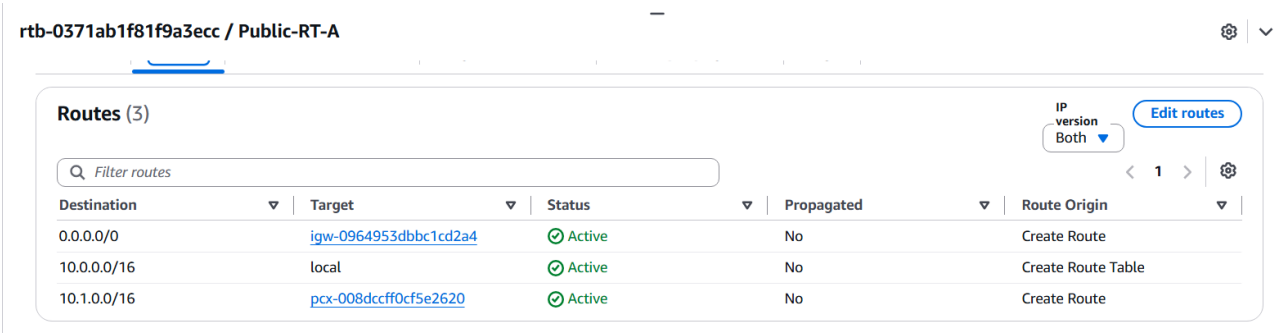


Figure 3. VPC-A private route table, including the route to VPC-B via peering.

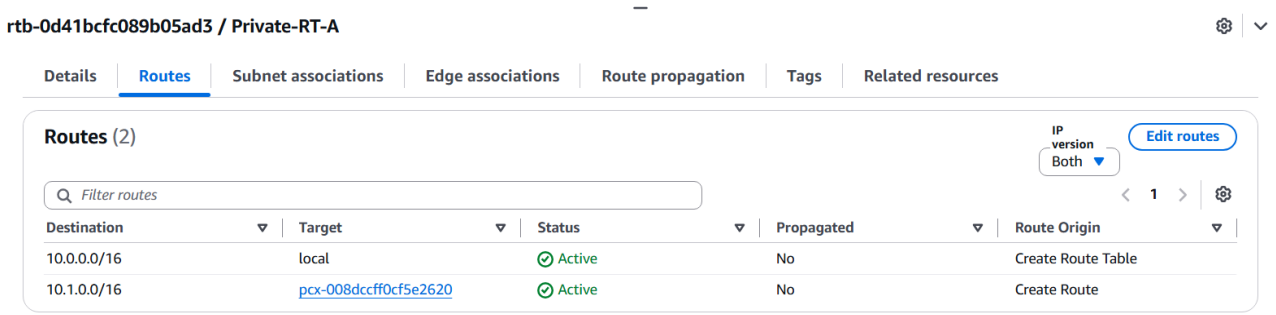


Figure 4. VPC-B private route table, including the return route to VPC-A via peering.

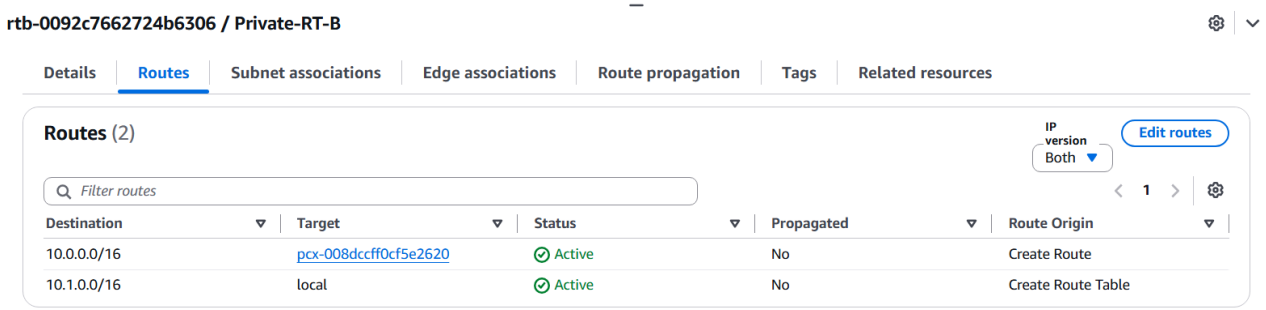


Figure 5. EC2 console showing Bastion-VPC-A and both private servers.

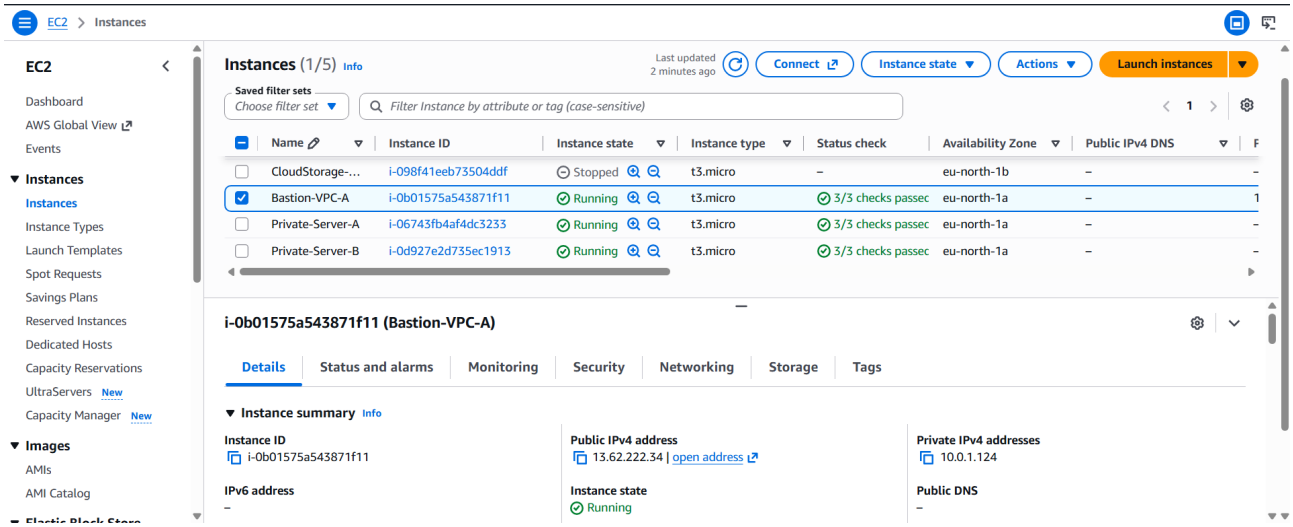


Figure 6. Successful ping test to 10.1.1.81 with 0% packet loss.

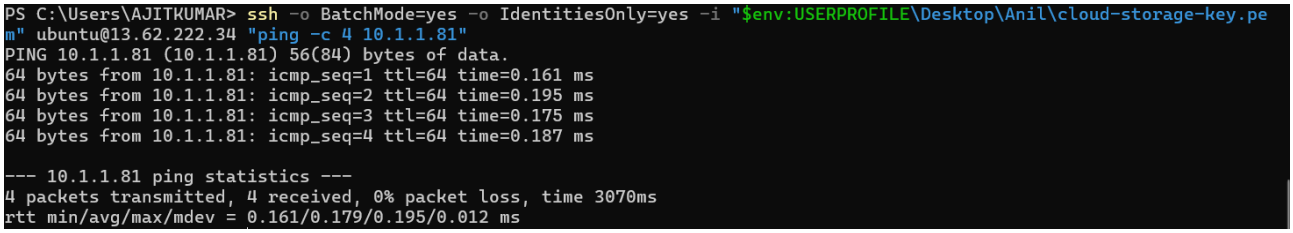


Figure 7. Successful SSH connection through the bastion to Private-Server-B.

